

ONT QC Suite

Plot Reference Guide

A complete description of every plot generated by **ont_qc.py**.
Includes what each visualisation shows, what to look for,
and how to interpret normal vs. abnormal findings.

Command	Description
<code>python ont_qc.py --runName MyRun</code>	Auto-detect all input files in the current directory
<code>python ont_qc.py --file summary.txt --runName MyRun</code>	Explicit summary file path
<code>... --poreActivity pore_activity.csv</code>	Explicit pore activity file (otherwise auto-detected)
<code>... --throughput throughput_data.csv</code>	Explicit throughput file (otherwise auto-detected)
<code>... --subsample 0.5</code>	Use 50% of reads — faster for very large files
<code>... --outdir /path/to/results</code>	Custom output directory (default: <runName>_qc/)
<code>... --maxLength 50000</code>	Cap read-length axis at 50 kb instead of mean + 3 SD
<code>... --prop 50000</code>	Change proportion-above-cutoff threshold (default: 20,000 bp)
<code>... --video</code>	Generate animated channel strand-time video (requires ffmpeg)
<code>python plot_guide.py</code>	Generate this PDF (requires: pip install reportlab)

Input File	Description	Required?
sequencing_summary*.txt	Per-read TSV from Albacore, Guppy, or Dorado	Yes
pore_activity*.csv	Per-minute channel state counts	Optional
throughput_*.csv	Per-minute cumulative counters	Optional

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1 Throughput

Pass-Filter Cumulative Yield vs. Time

Source: *sequencing_summary*.txt*

A scatter plot of pass-filter reads plotted cumulatively over run time. Each point represents a read; its x-position is the time it was sequenced (hours) and its y-position is the total yield accumulated up to that point (gigabases). The overall shape of the curve reflects how productively the flowcell performed throughout the run.

What to look for:

- Steepness of the early slope — a steep initial rise indicates a well-loaded, active flowcell.
- Changes in slope — shoulders or plateaux indicate periods of reduced output.
- Whether the curve levels off well before the run end.

How to interpret:

- A smooth S-curve (fast start, gradual plateau) is normal: pores deplete over time.
 - A very shallow early slope suggests poor loading, buffer issues, or a cold start — check active pores at T=0.
 - Sudden flat sections followed by recovery are often MUX scan events where channels are temporarily taken offline.
 - A curve that plateaus very early and never recovers may indicate a failed flowcell or a library prep issue.
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Reads per Hour

Source: *sequencing_summary*.txt*

A histogram binned by clock hour showing how many reads were produced in each hour of the run. Unlike the cumulative yield curve, this non-cumulative view makes productivity stalls immediately visible as short or absent bars.

What to look for:

- Hours with dramatically lower read counts — these are stalls.
- Whether read rate declines smoothly or drops abruptly.
- Any hours with zero or near-zero reads.

How to interpret:

- A gradual decline from left to right is expected as pores deplete.
 - Sharp drops followed by partial recovery are usually MUX scans.
 - A single empty hour early in the run may indicate the sequencer paused for a MUX scan or pore equilibration.
 - Sustained low output after a drop (no recovery) suggests pore death, a blockage event, or the flowcell running out of active pores.
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Yield per Hour [optional]

Source: *throughput_*.csv*

A bar chart of total basecalled yield (Gb) produced in each clock hour, derived from the throughput CSV. This complements "Reads per Hour" by accounting for read length — an hour with few but very long reads will still show high yield here.

What to look for:

- Consistency of yield across hours.
- Hours where yield drops without a corresponding drop in read count (indicates shorter reads).

How to interpret:

- Yield per hour naturally declines as pores deplete — this is normal.
 - A sudden drop in yield without a matching drop in read count means reads became shorter — check translocation speed and N50 over time.
 - Very high early yield that falls rapidly may indicate a large DNA fragment library being sequenced through quickly.
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2 Run Health over Time

Active Pores per Hour

Source: *sequencing_summary*.txt*

A bar chart showing the number of unique channels that produced at least one read in each clock hour. This is the primary indicator of flowcell health: as pores deplete, die, or become blocked, this number declines.

What to look for:

- The count at hour 0 — this is your effective flowcell loading.
- Rate of decline — slow and smooth vs. sudden drops.
- Any partial recoveries after drops (MUX scan fingerprint).

How to interpret:

- A high starting count (close to the flowcell maximum: 512 for MinION, 2675 for PromethION) indicates good loading.
 - Starting counts below ~50% of the flowcell maximum suggest poor loading, an aged flowcell, or a library too dilute or too concentrated.
 - Step-like drops then partial recoveries are MUX scans switching to fresh pore wells — this is normal and healthy.
 - A monotonic, steep decline without recoveries suggests irreversible pore fouling (common with dirty libraries or aged flowcells).
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Mean Q-Score over Time

Source: sequencing_summary*.txt

A line plot of mean Q-score per 10-minute bin (all reads, pass + fail) with ± 1 standard deviation shaded. Tracks whether basecall accuracy changes as the run progresses.

What to look for:

- Whether Q-score is stable, improving, or declining over time.
- Width of the shaded band — narrow = consistent quality.
- Early vs. late run quality difference.

How to interpret:

- A stable flat line at or above your threshold (Q10 for kit14, Q7 for older kits) is ideal.
 - A gradual decline is normal and expected as pores age; a steep early decline suggests reagent or temperature issues.
 - Sudden improvement mid-run can follow a MUX scan loading fresh pores with cleaner signal.
 - Very wide SD bands indicate heterogeneous quality — common with mixed-length or degraded libraries.
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Cumulative N50 over Time

Source: sequencing_summary*.txt

The N50 is recomputed cumulatively at the end of each clock hour and plotted as a line. N50 is the length such that 50% of all sequenced bases come from reads at least this long. Watching it evolve over time reveals whether fragment length is improving, degrading, or stable.

What to look for:

- Whether N50 rises, falls, or stabilises as the run proceeds.
- How quickly N50 stabilises — small runs may show large early fluctuations.

How to interpret:

- N50 typically rises in the first few hours as more reads accumulate and then plateaus once the dataset is statistically stable.
 - A steadily falling N50 late in the run means shorter fragments are being sequenced — long molecules may have been consumed or pore-blocking events are truncating reads.
 - A very low N50 that never rises suggests a degraded library; consider running a new prep.
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Translocation Speed over Time

Source: sequencing_summary*.txt

Mean translocation speed (bp/s) ± 1 SD per 10-minute bin, restricted to pass-filter reads. Translocation speed reflects how fast DNA moves through the nanopore and is sensitive to temperature, buffer conditions, and motor protein function.

What to look for:

- Baseline speed (typically 400–500 bp/s for R10.4 Kit14 at 400 bps mode).
- Drift in speed over time — upward or downward.
- Sudden shifts that correlate with changes in other metrics.

How to interpret:

- Speed is set by the motor protein and should be relatively stable for a given kit — minor drift of $\pm 10\%$ is normal.
 - A gradual speed increase often indicates a warming sequencer or evaporation concentrating the buffer.
 - Sudden speed drops may indicate the sequencer is struggling to keep up with data throughput (adaptive sampling backpressure) or a temperature drop.
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Pore Survival Curve

Source: *sequencing_summary*.txt*

Channels that produced a read in the first 10 minutes define the starting pool (100%). Each subsequent hour shows what fraction of those original pores are still active. This is the classic ONT pore survival metric — directly analogous to a cell survival curve in biology.

What to look for:

- Half-life of the pore population — the hour at which ~50% survive.
- Shape of the curve: exponential decay vs. abrupt steps vs. plateau.
- Final survival fraction at run end.

How to interpret:

- Smooth exponential decline is normal; individual pores burn out stochastically over the run.
 - Step-like drops aligned with MUX scans are expected — those pores were switched to new wells, so they may reappear briefly.
 - A very steep early drop (>50% loss in the first few hours) suggests pore fouling from dirty DNA, excessive adapter, or a library that is too concentrated.
 - A flat curve with >80% survival through the whole run indicates minimal pore loss — great for long-read sequencing.
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Sequencing Rate (Gb/hr) [optional]

Source: *throughput_*.csv*

Instantaneous output rate in Gb/hr derived by differencing adjacent rows in the throughput CSV. Shows the moment-to-moment productivity of the sequencer rather than cumulative totals.

What to look for:

- Peak rate — reached typically in the first 1–2 hours.
- Shape of decline — smooth or irregular.
- Spikes or troughs that correlate with events in other plots.

How to interpret:

- Rate declines naturally as pores deplete; this is expected.
 - Short spikes followed by drops are often MUX scans: a brief pause then burst of output from freshly selected pores.
 - Extended flat sections near zero before run end suggest the flowcell should have been stopped earlier.
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Read Acquisition Rate (reads/min) [optional]

Source: *throughput_*.csv*

Number of reads captured per minute, derived from the throughput CSV. Complements yield-based metrics: read rate is high when many short reads are captured, even if yield per read is low.

What to look for:

- Sustained high rates indicate good pore activity.
- Discordance between read rate and yield rate (could indicate short reads).

How to interpret:

- A high read rate with low yield suggests many short reads — check N50 and read length distribution.
 - A very low read rate despite many active pores may indicate the library is at low concentration and pores are idle between reads.
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Rolling Pass Rate over Time [optional]

Source: `throughput_*.csv`

30-minute rolling average of the fraction of reads that pass the quality filter (Q-score threshold), expressed as a percentage. A dashed reference line is drawn at 90%.

What to look for:

- Whether pass rate stays above the reference line.
- Trends: rising, falling, or stable.
- Whether pass rate correlates with changes in active pore count.

How to interpret:

- Pass rates consistently above 90% indicate high-quality signal; below 80% is a concern for most applications.
 - A declining pass rate over time may indicate degrading signal quality from ageing pores.
 - A very low pass rate early in the run (first 30 min) often normalises once pores equilibrate — watch for recovery.
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Estimated vs. Basecalled Bases (Cumulative) [optional]

Source: `throughput_*.csv`

Two cumulative lines plotted together: estimated bases (from raw signal length) and basecalled bases (reads that have been processed by the basecaller). When real-time basecalling keeps up with acquisition, the lines overlap. A growing gap indicates the basecaller is lagging behind.

What to look for:

- How closely the two lines track each other.
- Whether any gap closes later in the run (basecaller catching up).

How to interpret:

- Lines that overlap throughout indicate real-time basecalling running at full speed — ideal.
 - A widening gap means the basecaller cannot keep up with acquisition; the remaining reads will be basecalled post-run.
 - A gap that closes near the end of a run suggests the basecaller caught up after sequencing slowed down.
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3 Read Quality

Read Length Distribution

Source: *sequencing_summary*.txt*

A histogram of read lengths (all reads). The x-axis is capped at mean + 3 standard deviations to prevent extreme outliers from compressing the bulk of the distribution. A vertical dashed red line marks the N50.

What to look for:

- Shape: unimodal (one prep), bimodal (two size populations), or heavy right tail (ultra-long reads present).
- Position of the N50 relative to the distribution peak.
- Proportion of very short reads (< 1 kb) — often adapter dimers.

How to interpret:

- A roughly log-normal distribution peaked around 5–15 kb is typical for standard high-molecular-weight gDNA.
 - A large spike at very short lengths (<500 bp) may indicate carry-over of short DNA fragments, adapter dimers, or incomplete adapter ligation.
 - A bimodal distribution may indicate mixing of two library pools or incomplete size selection.
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Read Length Distribution — Pass vs. Fail (KDE)

Source: *sequencing_summary*.txt*

Overlapping kernel density estimate (KDE) curves of read lengths, with pass-filter reads in blue and fail reads in red. Both distributions are plotted on the same axes so that the length profiles of passing and failing reads can be compared directly. The x-axis is capped at mean + 3 SD of the pass-filter reads. A log x-axis is used when `--logLength` is set.

What to look for:

- Whether fail reads are systematically shorter than pass reads.
- Whether the two distributions overlap substantially or are well separated.
- A sharp fail peak at very short lengths (adapter dimers or short fragments).

How to interpret:

- Fail reads that are shorter on average than pass reads is expected: short fragments produce noisier signal that is more likely to fall below the Q-score threshold.
 - Near-identical pass and fail length distributions suggest that quality is not length-dependent — the fail reads may reflect pore noise or signal artefacts rather than short DNA.
 - A bimodal fail distribution (short peak + longer tail) may indicate two populations: adapter dimers and genuine reads that are just below the quality threshold.
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Read Length CDF (Base-Weighted)

Source: *sequencing_summary*.txt*

The cumulative distribution function of read lengths weighted by base count. The y-axis shows the fraction of total sequenced bases that come from reads up to a given length. N50 crosshairs are drawn: the curve crosses $y=0.5$ at exactly the N50 length by definition.

What to look for:

- Steepness: a steep curve means most bases are in long reads.
- Whether the curve reaches 1.0 within the displayed range (if not, very long reads extend beyond the axis cap).

How to interpret:

- A curve that rises slowly and reaches 0.5 at a high length value indicates a long-read-enriched library — desirable for most long-read applications.
 - A curve that reaches 0.5 at a low length value (low N50) suggests a short-fragment-dominated library.
 - The upper tail of the curve shows what fraction of total bases come from ultra-long reads.
-

Proportion of Bases by Read Length

Source: *sequencing_summary*.txt*

A scatter plot of read length (x) vs. cumulative proportion of total pass-filter bases (y), with reads sorted longest-first. This is equivalent to the inverse-sorted CDF and directly answers: "What fraction of my total yield comes from reads longer than X?"

What to look for:

- The x-value at $y=0.5$ is the N50.
- How flat the curve is at large x — a long flat tail means a useful population of ultra-long reads.

How to interpret:

- If the curve reaches 0.9 (90% of bases) at a short read length, the library is short-fragment dominated.
 - A curve with a long, slowly-declining tail beyond the N50 indicates a broad, well-distributed size range — typical of good HMW DNA extraction.
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Q-Score Distribution (Pass vs. Fail)

Source: *sequencing_summary*.txt*

Kernel density estimate (KDE) curves of per-read mean Q-score, with pass-filter reads in blue and fail reads in red, both overlaid. The pass/fail split is defined by the Q-score threshold set in MinKNOW (Q10 for Kit14, Q7 for older kits).

What to look for:

- Separation between pass and fail peaks.
- Width of the pass-filter peak — narrow is good (consistent quality).
- Whether any pass-filter reads have surprisingly low Q-scores.

How to interpret:

- Well-separated, narrow peaks indicate clean, high-quality data.
 - A broad pass peak suggests heterogeneous library quality — possibly degraded DNA or a mixture of fragment sizes.
 - A large fail peak overlapping the pass peak near the threshold is common and expected; reads near the threshold are borderline.
 - For kit14 data, the pass peak should be centred around Q15–Q20.
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Q-Score Tiers over Time

Source: *sequencing_summary*.txt*

Stacked area chart showing the proportion of pass-filter reads in each of four Q-score tiers (Q7–10, Q10–15, Q15–20, Q≥20) per clock hour. Colours grade from red (low Q) to green (high Q). Each column always sums to 100%.

What to look for:

- Whether the green (high Q) fraction increases or decreases over time.
- Any sudden shifts in the tier composition mid-run.

How to interpret:

- A stable tier distribution throughout the run indicates consistent basecall accuracy.
 - An increasing red fraction late in the run suggests pore ageing is degrading signal quality.
 - A sudden shift to higher Q-score tiers after a MUX scan reflects fresh, clean pores producing better signal.
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Read Length vs. Q-Score (Hex Joint Plot)

Source: *sequencing_summary*.txt*

A 2-D hexbin density plot of read length (x) against mean Q-score (y) for a random sample of up to 20,000 pass-filter reads. Hex colour encodes read density. Marginal distributions are shown on each axis.

What to look for:

- The location of the densest hexbin cluster — this is the modal read.
- Correlation between length and Q-score: does quality drop for longer reads?
- Any unexpected clusters (e.g., a separate population of low-Q, long reads).

How to interpret:

- A slight negative correlation between length and Q-score is normal: very long reads have more accumulated pore noise.
 - A strong negative correlation may indicate the motor protein is struggling with very long molecules.
 - Two distinct clusters may suggest a mixed library (e.g., two PCR products of different sizes).
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Read Length Distribution by Q-Score Tier (Violin Plot)

Source: *sequencing_summary*.txt*

Violin plots — one per Q-score tier — showing the full distribution of pass-filter read lengths within each quality band. An inner box-and-whisker plot marks the median and IQR. Colours grade red-to-green from low to high Q.

What to look for:

- Whether the median read length decreases as Q-score tier increases.
- Width of the violin — wide = many reads; narrow = few.
- Any tier that is entirely absent or extremely narrow.

How to interpret:

- A consistent read length across all tiers means quality does not depend on fragment length for this library.
 - A clear pattern where high-Q reads are short indicates the library has excellent short-read basecalling but long reads are noisier.
 - An empty or near-empty Q≥20 tier is expected for R9.4.1 data; for R10.4 Kit14 data this tier should be well populated.
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Read Length Distribution by Q-Score Tier (KDE)

Source: *sequencing_summary*.txt*

Overlapping, filled KDE curves — one per Q-score tier — of pass-filter read lengths plotted on a log x-axis. Tiers are Q7–10 (red), Q10–15 (yellow), Q15–20 (light green), Q≥20 (dark green), matching the colours of the violin plot. Vertical dashed lines mark the median length for each tier. Unlike the violin plot, this view shows distributional overlap between tiers directly and handles extreme outliers more gracefully on the log scale.

What to look for:

- Whether the KDE peaks for each tier align or are shifted relative to each other.
- The spread of each curve — narrow peaks indicate a consistent length profile within that quality tier.
- Tiers with very flat or absent curves (few reads in that quality band).

How to interpret:

- Peaks that are well aligned across tiers indicate read length and quality are independent for this library.
 - A leftward shift of the low-Q (red) tier indicates that shorter reads are more likely to fail — common with adapter carry-over or fragmented DNA.
 - A rightward shift of the high-Q (green) tier indicates that longer reads basecall better — expected for libraries with good high-molecular-weight DNA.
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Read End Reason

Source: *sequencing_summary*.txt*

A horizontal bar chart showing the count of reads grouped by the reason the sequencer stopped reading them. Each bar is labelled with its exact count. Common end reasons are: `signal_positive` (read completed normally), `signal_negative` (pore blocked), `mux_change` (taken off by MUX scan), `unblock_mux_change` (rejected by adaptive sampling), and `data_service_unblock_mux_change` (rejected by a data service rule). This plot is only generated when the `end_reason` column is present.

What to look for:

- Which end reason dominates.
- The proportion of `unblock_mux_change` events — these are adaptive sampling rejections.
- The proportion of `signal_negative` events — these are blockages.

How to interpret:

- `signal_positive` should be the most common end reason for a normal run.
 - High `unblock_mux_change` is expected and desirable on adaptive sampling runs; on non-adaptive runs it may indicate the library contains many short fragments that are being forcibly ejected.
 - High `signal_negative` (blockages) can indicate dirty DNA, excess adapter, or pores that are becoming clogged.
 - High `mux_change` counts correlate with MUX scan events — normal.
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Read Length vs. Run Time (Hexbin)

Source: *sequencing_summary*.txt*

A 2-D hexbin density plot showing how read length (y) is distributed across run time (x), restricted to pass-filter reads. Reveals whether the fragment length composition changes as the run progresses.

What to look for:

- Whether the length distribution shifts over time.
- Diagonal banding — often indicates a correlated relationship.
- Any time windows where the distribution collapses to very short reads.

How to interpret:

- A stable, vertically symmetric blob throughout the run indicates no time-dependent length bias.
 - A gradual compression toward shorter reads late in the run is common: long fragments are consumed first by the most active pores.
 - Very short reads late in the run with no long reads may indicate that long fragments have been fully sequenced and only short DNA remnants remain.
-

Sequencing Speed over Time (Scatter)

Source: *sequencing_summary*.txt*

All reads (pass and fail) plotted as points with start time on x and sequencing speed (bp/s) on y. Pass-filter reads are blue, fail reads are red. A random subsample of up to 50,000 reads is shown to keep the plot tractable for large runs.

What to look for:

- The modal speed band — is it where you expect for your kit?
- Outlier reads with extreme high or low speed.
- Any time-dependent drift in the speed distribution.

How to interpret:

- R10.4 Kit14 400 bps mode → expected speed ~400 bp/s.
 - R9.4.1 450 bps mode → expected speed ~450 bp/s.
 - Very fast reads (>1000 bp/s) are often noise or very short fragments that zip through the pore without the motor protein slowing them down.
 - Very slow reads (<100 bp/s) may indicate a stalling motor protein or a difficult DNA secondary structure.
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Median Current over Time [optional]

Source: *sequencing_summary*.txt*

Scatter plot of per-read median ion current (pA) vs. start time for all reads (pass and fail). The median template current reflects the electrical environment of the nanopore. This plot is only generated when the `median_template` column is present (older files).

What to look for:

- Drift in the median current over the run.
- Whether fail reads cluster at systematically different current values.

How to interpret:

- A stable median current around 60–80 pA is typical for most kits and flowcells.
 - A gradual upward drift may indicate buffer evaporation concentrating the salt solution and increasing conductivity.
 - A large spread between pass and fail median currents suggests the current threshold is important for quality classification.
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4 Flowcell Channel Map

Flowcell Channel Map (3-Panel Heatmap)

Source: *sequencing_summary*.txt*

Three stacked heatmaps, each showing a property per physical channel on the flowcell membrane. The grid matches the MinKNOW channel layout: 16×32 for MinION/GridION (512 channels) or 25×120 for PromethION (3,000 channels). Only pass-filter reads contribute. Panel 1: mean Q-score. Panel 2: read count. Panel 3: yield (Mb).

What to look for:

- Spatial uniformity — should be roughly even across the flowcell.
- Any dark (dead) patches — rows, columns, or quadrants with no activity.
- Rows or columns with unusually high or low Q-score relative to neighbours.

How to interpret:

- Uniform colour across all three panels indicates even loading and uniform membrane quality — the ideal outcome.
- An entire row of dark channels often indicates a bad connection or a failed electrode group.
- A quadrant of dead channels corresponds to one of the four MUX groups failing — check the Pore Activity plot for that group.
- High read count but low yield in a region indicates short reads from that part of the membrane.
- Low Q-score in one corner may indicate uneven temperature distribution or physical contamination on the membrane surface.

Pass-Filter Yield by Flowcell Region

Source: *sequencing_summary*.txt*

A bar chart comparing total pass-filter yield (Gb) across flowcell regions. For MinION/GridION, channels are divided into 4 groups of 128 channels each. For PromethION, 12 blocks of 250 channels. Bars are ordered numerically by region number and coloured blue.

What to look for:

- Uniformity of bar heights — large disparities indicate regional problems.
- One or more regions with zero or near-zero yield.

How to interpret:

- Roughly equal bars across all regions indicate uniform flowcell loading and performance.
- One region significantly below others may indicate that MUX group had fewer active pores, possibly due to a loading artefact or a section of membrane that was damaged or air-exposed.
- For PromethION, large inter-block variation is common due to the physical distance between the blocks on the chip.

5 Barcode Analysis [barcoded runs only]

These plots are generated automatically when the sequencing summary file contains a `barcode_arrangement` column. Unclassified reads are excluded from all barcode plots by default.

Read Counts per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Horizontal bar chart of pass-filter read counts for each classified barcode, sorted alphabetically. Provides a quick visual balance check across samples.

What to look for:

- Balance between barcodes — are any samples dramatically under- or over-represented?
- Any barcodes with zero reads.

How to interpret:

- For balanced multiplexing, bars should be roughly equal height.
 - A factor of 2–3× variation is common and acceptable; 10× variation indicates a pipetting error or very different DNA concentrations in the pooled library.
 - Zero reads for a barcode means that sample was not detected at all — check the library preparation for that sample.
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Yield per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Horizontal bar chart of total pass-filter yield (Mb) per barcode. Combines read count and read length — a barcode with fewer but longer reads may have the same yield as one with many short reads.

What to look for:

- Which barcodes achieved coverage targets.
- Discordance between read count and yield bars for the same barcode.

How to interpret:

- A barcode with low read count but normal yield has longer reads — check the read length violin plot.
 - A barcode with high read count but low yield has unusually short reads — check for sample degradation or adapter contamination.
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Q-Score Distribution per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Violin plot with one violin per barcode (width-normalised) showing the distribution of mean Q-scores for pass-filter reads. An inner box-and-whisker plot marks the median and IQR.

What to look for:

- Any barcode with consistently lower Q-score than the others.
- Unusually wide violins indicating high Q-score variance in one sample.

How to interpret:

- Most barcodes should have similar Q-score distributions if the multiplexed libraries had similar fragment size and purity.
 - A single barcode with low Q may indicate that sample had PCR inhibitors, damaged DNA, or a different GC content.
-

Read Length Distribution per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Violin plot with one violin per barcode showing the distribution of pass-filter read lengths. Lengths are capped at mean + 3 SD to prevent extreme outliers from collapsing all violins.

What to look for:

- Whether all barcodes have similar size distributions.
- Any barcode with a very different modal length.

How to interpret:

- If barcodes were prepared from samples of similar size, violins should look nearly identical.
 - A barcode with a much lower median length may represent a degraded DNA sample or a library with excess short-fragment contamination.
 - A much longer median for one barcode may indicate that sample had a larger insert size or was not size-selected.
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Read Length CDF per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Base-weighted cumulative distribution function (CDF) of read length for each barcode — one coloured line per barcode. A dashed grey line marks $y=0.5$ (the N50 level). The x-axis is capped at mean + 3 SD.

What to look for:

- How close together the lines are — separation indicates different size distributions.
- Where each line crosses $y=0.5$ — this is the per-barcode N50.

How to interpret:

- Tightly clustered lines indicate reproducible size selection across samples in the same pool.
 - A line shifted far left (early $y=0.5$ crossing) indicates that barcode has shorter reads — check the violin plot and Q-score plot for corroborating evidence.
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Cumulative Read Accumulation per Barcode [optional]

Source: *sequencing_summary*.txt* (barcoded)

Stacked area chart of cumulative read count per barcode over run time (one colour per barcode). Shows when each sample was sequenced and whether yield accumulation was consistent across the run.

What to look for:

- Whether all barcodes accumulate reads at similar rates.
- Any barcode that stops accumulating reads mid-run.
- Whether proportions stay stable or shift over the run.

How to interpret:

- Parallel, evenly-spaced accumulation lines indicate balanced sequencing of all samples throughout the run.
 - A barcode that plateaus early relative to others may represent a sample with fewer molecules in the pool.
 - Changing proportions over time may reflect different fragment length distributions competing for pore occupancy.
-

6 Pore Activity / Duty Time [requires pore_activity*.csv]

These plots require the *pore_activity*.csv* file produced by MinKNOW. This file records the count of channels in each state at each minute of the run.

Pore Activity / Duty Time (Interactive) [optional]

Source: *pore_activity*.csv*

An interactive Plotly stacked bar chart showing how many channels are in each state at each minute of the run. States include: strand (actively sequencing), pore (open, available), adapter (reading adapter sequence), unblocking (ejecting a molecule), no_pore, locked, saturated, and others. Hover over any bar to see exact counts. Toggle states on/off by clicking the legend. A static PNG version is also generated as a fallback.

What to look for:

- The "strand" state (blue) — this is time spent actively sequencing.
- The "no_pore" state (grey) — channels with no functional pore.
- The "locked" state (brown) — pores stuck and unable to sequence.
- Sudden increases in "saturated" (red) state.

How to interpret:

- High "strand" fraction relative to total channels = high duty time = efficient sequencing.
 - Increasing "no_pore" over time is expected as pores deplete — these channels will not recover without a fresh flowcell.
 - A surge in "locked" state may indicate DNA or protein contamination blocking the pore entrance.
 - High "saturated" counts (positive current overload) can occur with very concentrated libraries or if the flowcell is too hot.
 - Periodic spikes in "unblocking" state are MUX scan events ejecting molecules from all active channels simultaneously.
-

Pore Occupancy over Time [optional]

Source: *pore_activity*.csv*

Line plot (with shaded area) of pore occupancy (%) over run time. Occupancy is defined as: $\text{strand} \div (\text{strand} + \text{pore} + \text{adapter} + \text{unblocking}) \times 100$. The denominator counts only channels that have an active pore — so occupancy reflects how productively the available pores are being used, independent of how many pores have died.

What to look for:

- Baseline occupancy — is it consistently high?
- Whether occupancy declines, rises, or fluctuates over the run.
- Periodic dips corresponding to MUX scans.

How to interpret:

- Occupancy above 70% indicates a library at good concentration: available pores spend most of their time on strands.
 - Occupancy below 30% suggests the library is too dilute — pores spend most of their time open and waiting for a molecule to arrive.
 - Occupancy above 95% consistently may indicate the library is too concentrated, causing strand queuing and premature stalls.
 - Occupancy that declines over time may indicate library depletion as molecules are consumed.
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Blocked vs. Active Channels over Time [optional]

Source: *pore_activity*.csv*

A stacked area chart that collapses the ~15 raw MinkNOW channel states into four interpretable groups, expressed as a percentage of all channels: Sequencing (strand — DNA actively translocating), Ready (pore, adapter, unblocking — available and about to sequence), Blocked (locked, saturated, multiple, pending_manual_reset — stuck or occupied unproductively), and Unavailable (no_pore, zero, disabled, and all remaining states). This simplified view makes it immediately apparent whether blocked pores are accumulating, whether available pores are being used productively, and how the ratio shifts across the run.

What to look for:

- The combined height of Sequencing + Ready (blue + green) at T=0 -- this is the effective active-channel fraction at run start.
- Whether the Blocked (orange) band grows over time.
- Whether the Unavailable (grey) band grows steadily (normal pore depletion) or in sudden steps (MUX scan events).
- Any recovery in the Sequencing band after a MUX scan.

How to interpret:

- A tall Sequencing + Ready band at T=0 (>60% of channels) indicates a well-loaded flowcell with good initial pore availability.
 - Gradual growth of the Unavailable band throughout the run is normal and reflects irreversible pore loss as the flowcell ages.
 - Growth of the Blocked band over time suggests progressive pore fouling -- common when the library contains free adapters, protein carry-over, or inhibitors from the sample preparation.
 - Periodic spikes where Sequencing drops and Unavailable rises briefly are MUX scan fingerprints -- expected and healthy.
 - A Blocked band that rises steeply in the first few hours on a fresh flowcell strongly suggests a library quality issue rather than flowcell degradation.
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7 Top 10 Longest Pass-Filter Reads

Top 10 Longest Pass-Filter Reads Table

Source: *sequencing_summary*.txt*

A table listing the 10 longest pass-filter reads by read length (bp), including each read's unique identifier (read_id) and length in base pairs. Reads are ranked 1 (longest) to 10.

What to look for:

- The length of the longest read — is it consistent with the expected maximum fragment size for your extraction method?
- Large gaps between consecutive entries in the list.

How to interpret:

- Read IDs can be used to extract specific reads from the FASTQ/BAM for manual inspection: `samtools view -h file.bam`.
 - A longest read of 500 kb or more indicates excellent ultra-long DNA extraction; above 1 Mb is exceptional.
 - If the longest read is only 20–30 kb, the library lacks very long fragments — consider switching to an ultra-long extraction protocol.
 - Very similar lengths among the top 10 (e.g., all around 200 kb) may indicate a gel plug extraction with consistent fragment size.
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